Trigonometric Limits, Derivatives, and Inverse Trigonometric Examples

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§1.1 Trigonometric limits

The note collects examples about trigonometric functions and derivatives. The first basic limit is

$$\lim_{x\to 0}\frac{\sin x}{x}=1.$$

From this one obtains

$$\lim_{x\to 0}\frac{1-\cos x}{x}=0,\qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}.$$

The second follows from

$$1-\cos x=2\sin^2\frac{x}{2}.$$

The handwritten page uses these limits repeatedly when deriving derivatives of trigonometric functions.

§1.2 Derivative of sine and cosine

Using the addition formula,

$$\sin(x+h)-\sin x=\sin x(\cos h-1)+\cos x\sin h.$$

Therefore

$$\frac{\sin(x+h)-\sin x}{h}=\sin x\frac{\cos h-1}{h}+\cos x\frac{\sin h}{h}.$$

Letting $h\to 0$ gives

$$(\sin x)'=\cos x.$$

Similarly,

$$\cos(x+h)-\cos x=\cos x(\cos h-1)-\sin x\sin h,$$

so

$$(\cos x)'=-\sin x.$$

§1.3 Tangent, cotangent, secant, and cosecant

Using quotient rule,

$$\tan x=\frac{\sin x}{\cos x}\quad\Longrightarrow\quad(\tan x)'=\frac{1}{\cos^2 x}=\sec^2 x.$$

Similarly,

$$(\cot x)'=-\csc^2 x,\qquad(\sec x)'=\sec x\tan x,\qquad(\csc x)'=-\csc x\cot x.$$

The note includes examples where the algebra is simplified by rewriting everything in sine and cosine first.

§1.4 Inverse trigonometric derivatives

For $y = \arcsin x$, we have

$$x = \sin y, \qquad 1 = \cos y\, y'.$$

Since $y \in [-\pi/2, \pi/2]$, $\cos y = \sqrt{1 - x^2}$, so

$$(\arcsin x)' = \frac{1}{\sqrt{1 - x^2}}.$$

Similarly,

$$(\arccos x)' = -\frac{1}{\sqrt{1 - x^2}}, \qquad (\arctan x)' = \frac{1}{1 + x^2}.$$

The page also works through reciprocal-trigonometric inverse derivatives, using triangles to determine the missing side length.

§1.5 Examples from the handwritten page

A typical example differentiates a composite expression such as

$$y = \frac{\sin x}{1 + \cos x}.$$

One can use quotient rule directly, but the note also points out the simplification

$$\frac{\sin x}{1 + \cos x} = \tan \frac{x}{2}.$$

Thus

$$y' = \frac{1}{2} \sec^2 \frac{x}{2}.$$

This illustrates a useful habit: before differentiating, look for a trigonometric identity that simplifies the expression.

Another example uses implicit differentiation for equations involving $\sin x$ and $\cos x$. The method is always the same: differentiate both sides, then solve for the desired derivative.

§1.6 A closing lens

The derivatives of trigonometric functions are not isolated formulas. They all come from the same two sources: the fundamental limit $\sin h/h \to 1$ and the addition formulas. Inverse trigonometric derivatives then come from implicit differentiation and right-triangle identities. The note is mainly about learning to move between these representations fluently.

§1.7 The circle as a Lie group

The unit circle can be written as

$$S^1 = \{e^{it} : t \in \mathbb{R}\}.$$

Multiplication of complex numbers adds angles:

$$e^{is} e^{it} = e^{i(s+t)}.$$

Thus trigonometric addition formulas are group laws in coordinates. The identities for sine and cosine are the real and imaginary parts of this multiplication rule.

§1.8 Trigonometric derivatives from the exponential

Euler's formula

$$e^{ix} = \cos x + i \sin x$$

turns differentiation of trigonometric functions into differentiation of e^{ix}:

$$\frac{d}{dx} e^{ix} = ie^{ix} = i \cos x - \sin x.$$

Comparing real and imaginary parts gives

$$(\cos x)' = -\sin x, \qquad (\sin x)' = \cos x.$$

§1.9 Inverse trigonometric branches

Inverse trigonometric functions require branch choices. For arcsin, sine is restricted to $[-\pi/2, \pi/2]$ so that it is one-to-one. The derivative

$$(\arcsin x)' = \frac{1}{\sqrt{1 - x^2}}$$

blows up at ± 1 because the original sine curve has horizontal tangent there.

§1.10 Trigonometry from circle symmetry and branches

Trigonometric limits and derivatives are not isolated formulas. They come from the geometry and group structure of the circle, while inverse functions require branch choices and the inverse function theorem.